

A trajectory simplification method based on the clustering of vessel speed and course segments

1st Jia Li
Navigation College
Dalian Maritime University
Dalian, China
lijia19962311@163.com

2nd Yongjie Zhu
Navigation College
Dalian Maritime University
Dalian, China
zyongjiedlmu@163.com

3rd Yi Zuo
Navigation College
Dalian Maritime University
Dalian, China
zuo@dlmu.edu.cn

4th Tieshan Li
Navigation College
Dalian Maritime University
Dalian, China
tieshanli@126.com

Abstract—Automatic Identification Systems (AIS) can provide massive ship trajectory data that is valuable for mining information in water traffic. However, it is also easy to cause the data redundancy, adding difficulty to the data processing. To address this problem, this paper proposes a trajectory simplification method based on the segmental clustering of the ship's speed and course. Firstly, the maneuver points of a trajectory are determined by the speed and course threshold. The trajectory is divided into segments in accordance with the maneuver points. Then, the feature points of each segment are extracted based on K-medoids clustering method. The maneuver points and feature points of every trajectory segments are combined as trajectory features to achieve the purpose of simplifying the ship's trajectory. The experiment results show that the method can well maintain the feature of original trajectory and eliminate the redundant track points in AIS trajectory. Compared with tradition Douglas-Peucker (DP) algorithm, the distortion rate could be reduced 0.2% with the same simplification.

Keywords—Automatic identification system (AIS), ship trajectory, trajectory compression, K-medoids algorithm, trajectory segments

I. INTRODUCTION

The ship automatic identification system (AIS) working in the very high frequency (VHF) band uses the time division multiple access technology. It could automatically broadcast and receive the ship static information, dynamic information, voyage-related information and safety information, identify and locate ships by exchanging data with nearby ships and AIS base stations. With the rapid development of the AIS terminal network, research on AIS data mining and analysis is increasing. Mou et al. (2010) used AIS data for ship collision avoidance studies in busy shipping lanes. By analyzing 62 days of data from the North Sea in June, July and August 2007, important parameters such as the closest point (CPA) and the time to reach the CPA (TCPA) were calculated.[1]Giuliana P et al.(2013) propose an unsupervised incremental learning based method for extracting ship motion patterns. Transformation of large amounts of AIS data into information to support decision making.[2] Altan(2018) had developed a simulator that simulates AIS communication for real or simulated marine traffic flow considering the movement of each ship. AIS communication is evaluated

under various AIS transmitting conditions and for various aspects using thus developed AIS Simulator.[3]

All studies based on AIS data basically require the pre-processing of a large number of historical records. The ship sends AIS information at a period of 3-10 seconds. While, the actual ship's course and speed change frequency are much lower than the AIS transmission frequency. Therefore, most of the AIS trajectory data is redundant, which means little for describing the motion of the ship. Moreover, large amounts of AIS data has great influences on the data storage, transmission, processing and calculation. Therefore, it is essential and important to remove redundancy trajectory data from the original AIS data[4].

Patroumpas et al. (2017) proposed an AIS position-extracting trajectory simplification system, which preserved only the most recent significant motion characteristics by employing an online tracking module to detect significant changes in each ship's trajectory over time. By identifying such transient and long-term trajectory motion events such as stop, turn, or slow motion, a "critical point" of each ship's trajectory can be identified and provide a concise and reliable heading representation[5]. Le Qi (2016) proposed a ship trajectory data compression algorithm based on the identification of ship course changes. Since the variation of a ship's course of the ship's track was similar to the corner point on a straight line. The corner point detection algorithm was adopted to identify the variation of the ship's course[6]. The Douglas-Peucker (DP) algorithm was currently recognized as one of the most accurate and effective trajectory data compression methods, and there have been many related studies on the DP algorithm by scholars both at home and overseas[7]. Li et al proposed a method that considered the time attribute of ship trajectory. They considered a ship trajectory as the line data in three-dimensional space. The experimental results based on 30,000 records indicated that the method can retain additional feature points and preserve more information about the speed of trajectory.[8]. Guoyou Shi (2018) et al. proposed a method that combines the course variation of the ship's trajectory with the DP algorithm, which could significantly reduce the running time of trajectory data compression algorithm and improve the quality of the compressed data at a high compression strength. In addition, the ship trajectory data (based on a geo-coordinate system) was made directly available to the DP algorithm (based on a

Cartesian coordinate system) by using Mercator projection coordinates[9].

However, previous studies have not considered the quality of compressed data when the trajectory simplification intensity is high. In order to maintain the quality of the data even under high simplification intensity and the simplified trajectory can retain the main traffic flow characteristics, a trajectory simplification method based on the clustering of vessel speed and course segments was proposed in this paper. Firstly, the maneuver points of the ship's trajectory are determined in accordance with the ship's speed and course thresholds. Then the trajectory was segmented according to the maneuver points and the K-medoids clustering method is used to extract the features of the segmented trajectory. Finally, the features and maneuver points were combined together to form a new collection of trajectory points to effectively delete other redundant points. The proposed method was compared with tradition DP algorithm and the experiment results showed that the compression quality was improved with the same simplification rate.

The remaining components of the paper are structured as follows: Section 2 introduces the methodology. Section 3 shows experiment results and discussions and Section 4 is conclusions.

II. METHODOLOGY

A. Trajectory Segmentation

In this paper, the ship trajectory features are extracted based on the ship speed and course segments. We analyzed the trajectory characteristics and found that the ship finished maneuver operation mainly by changing course or speed. Therefore, in a trajectory, the points with large changes in course or speed are considered to be the maneuver points of the ship's trajectory for maneuvering or agitating. The trajectory segments between the two adjacent maneuver points belong to the trajectory point with little change in course and speed. The maneuver state of these points is more stable, which are considered to be stable points. Therefore, in the process of finding maneuver points, we only need to remain the points whose changing magnitude of course and speed is larger than the threshold. Moreover, the course and speed threshold are the parameters that should be determined in advance.

The trajectory information based on AIS in this paper mainly consists of four parts, namely, longitude (Lont), latitude (Lat), heading (c) and speed (v) as shown in Equation (1).

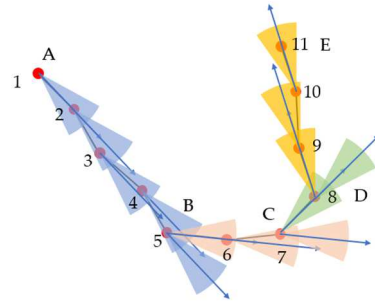
$$T = \{(Lat_i, Lont_i, c_i, v_i) \mid i=1,2,3,\dots,m\} \quad (1)$$

$$f(c_i, c) = \begin{cases} 1, & |c_i - c_{i-1}| \geq c \\ 0, & |c_i - c_{i-1}| < c \end{cases} \quad i = 2, 3, \dots, m \quad (2)$$

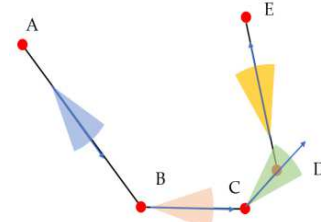
$$f(v_i, v) = \begin{cases} 1, & |v_i - v_{i-1}| \geq v \\ 0, & |v_i - v_{i-1}| < v \end{cases} \quad i = 2, 3, \dots, m \quad (3)$$

As shown in Equations (2) and (3), for trajectory point i , if $f(c_i)=1$ or $f(v_i)=1$, this trajectory point is considered to be a

maneuver point. While, if $f(c_i)=0$ and $f(v_i)=0$, this trajectory point is considered to be a stable point. Provided that there is a ship trajectory as shown in Fig.1.



(a) Original trajectory of ship



(b) Simplified trajectory

Fig. 1. Individual ship track

The red color is the AIS ship track point, assuming that the track point is numbered from 1 to 11. Among them, trajectory point 1 is the start point of this trajectory, and trajectory point 11 is the end point of the trajectory. Then the process of generating maneuver points is laid out as follows:

- (a) Initialize the set of maneuver point set Fea to an empty set.
- (b) Place trajectory start point into maneuver point set Fea as feature point 1.
- (c) Calculate the course and speed difference between trajectory point 2 and feature point 1 (trajectory point 1 in Fig.1). If the course or speed difference is greater than the set course or speed deviation threshold, place this trajectory point into maneuver point set Fea. Otherwise, continue to assess the next trajectory point.
- (d) Repeat step (c) until the last trajectory point is assessed.

B. Feature Extraction of Trajectory Segments Based on K-medoids Clustering Algorithm

According to the above extraction methods of maneuver points, we can obtain the maneuver points as A, B, C, D and E of the trajectory in Figure 1. By connecting these maneuver points in time sequence order, we can basically describe the shape of this trajectory. However, for the trajectory points located between two adjacent trajectory feature points, the changing magnitude in course and speed is relatively small. The prementioned maneuver point extraction method could not effectively capture the characteristics of this kind of trajectory points. Therefore, for extracting the characteristics of these trajectory points, the K-medoids clustering algorithm was adopted.

As shown in Fig.1, for trajectory points 1 to 5, there is also a slight change in course for this trajectory. Therefore, we applied the K-medoids clustering methods for the trajectory points located between two adjacent maneuver points, and adopted cluster centers as the trajectory feature point of trajectory segments. Then, the maneuver points and trajectory feature points of segments were arranged in accordance with the time sequence to form the feature of this trajectory [10].

The pseudo-code of proposed trajectory feature extraction method can be illuminated as follows:

Algorithm 1: Trajectory feature extraction
Input: P (trajectory point), C_tol (course deviation threshold), V_tol (speed deviation threshold), Fea (trajectory feature point set). Feature (start_point, end_point) means extracting the feature between these two adjacent maneuver points with K-medoids clustering methods. Output: trajectory feature point
Initialize: F $\rightarrow\emptyset$ Number of trajectory point N Fea $\leftarrow$ Fea.append(P ₁) sample_point=Fea ₁ 1: Divide the trajectory into segments in accordance with course and speed tolerance For i=2 to N do 2: if P _i ^(C) -sample_point ^(C) $\geq$ C_tol 3: Fea $\leftarrow$ Fea.append(P _i) 4: sample_point=P _i continue end if P _i ^(V) -sample_point ^(V) $\geq$ V_tol 5: Fea $\leftarrow$ Fea.append(P _i) 6: sample_point=P _i continue end end 7: Extract the feature between two adjacent points in Fea 8: Number of points in F M For j=1 to M-1 do 9: start_point=Fea _j 10: end_point=Fea _{j+1} 11: Fea $\leftarrow$ Fea.append(Feature (start_point, end_point)) end 12: Rank the trajectory feature point set F on the basis of time sequence

III. EXPERIMENTAL ANALYSIS AND RESULTS

A. Data Description

AIS messages include static, dynamic, and navigation-related messages. The information used in this paper are mainly dynamic messages. The contents of dynamic message are shown in Table I.

Data collection took place in the Port of Dalian from January 2015 to February 2015. The study area is shown in Fig.2 and Table II. We selected all AIS messages as the data

source and used the Maritime Mobile Service Identification (MMSI) number to identify the trajectory point of all vessels in the study area. The detail information of the experiment data is shown in Table III.

TABLE I. CONTENTS OF DYNAMIC MESSAGES

CONTENTS	DESCRIPTION
NAVIGATIONAL STATUS	Status including under way using engine and at anchor
SOG	Speed over ground
LONGITUDE	POSITION INFORMATION
LATITUDE	POSITION INFORMATION
COG	COURSE OVER GROUND
TH	TRUE HEADING

TABLE II. RANGE OF STUDY

BOUNDARY POINT	LONGITUDE
LEFT-BOTTOM	121° 47.16' E
RIGHT-TOP	121° 41.33' E

TABLE III. SIZE OF EXPERIMENTAL DATA

NUMBER OF POSITION MESSAGES	NUMBER OF SHIPS
19766	15

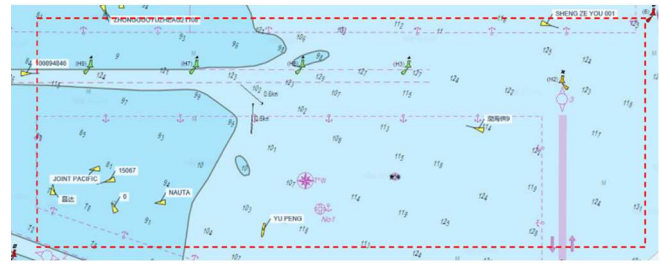


Fig. 2. Individual ship track

B. Evaluation Criteria

In this study, we evaluate the compression performance in terms of both distortion rate and simplification rate. Simplification rate is used to characterize the trajectory compression strength. Distortion rate is used to characterize the quality of compression. Therefore, these two evaluation criteria to assess the DP algorithm and our method [11-12].

1) Simplification Rates

In this paper, the simplification rate was defined as the ratio between the number of feature points after trajectory simplification and the total number of original trajectory points as shown in Equation (4).

$$s = \frac{m - n}{m} \% \quad (4)$$

The m is the total number of trajectory points of original trajectory and n stands for the number of trajectory points of simplified trajectory with our methods.

2) Distortion Rates

The trajectory feature points are arranged in chronological order and could be considered as time series data. Moreover, the number of trajectory points before and after simplification are not the same. The traditional ways of similarity calculation, such as Euclidean distance. Therefore, the dynamic time wrapping (DTW) was adopted here. The DTW could measure the similarity or dissimilarity of different number of time sequence. The value of DTW was defined as distortion rate in this paper. The lower distortion rate means that the method could better maintain the characteristics of original trajectory after simplification.

The theory of DTW can be interpreted as shown in Fig 3. The top and bottom two solid lines represent the two time series, and the dashed line between the time series represents the similar points between two time sequences. DTW uses the sum of the distances between all these similar points, called Warp Path Distance, to measure the similarity between the two time sequences[13].

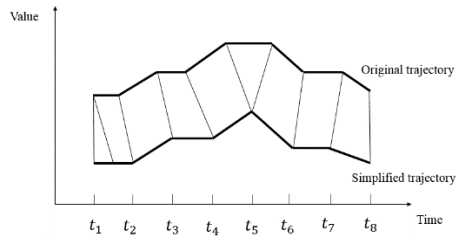


Fig. 3. Individual ship track

We define the wrap path of form $w = w_1, w_2, \dots, w_3$, $Max(|X|, |Y|) \leq K \leq |X| + |Y|$. But this path is not chosen arbitrarily. The normalized path must begin at $W_1 = (1, 1)$ and end at $W_k = (|X|, |Y|)$ to ensure that each coordinate of the and appears in the middle. In addition, the points must be added monotonically to ensure that the dashed lines in Figure 3 were monotonous as shown in Equation (5).

$$w_k = (i, j), w_{k+1} = (i', j') \quad i \leq i' \leq i+1, j \leq j' \leq j+1 \quad (4)$$

The final Warp path to be obtained is the one with the shortest distance to it as shown in Equation (6).

$$D(i, j) = Dist(i, j) + \min[D(i-1, j), D(i, j-1), D(i-1, j-1)] \quad (4)$$

C. Experimental Results

In order to verify the effectiveness of the method, the method is compared with the DP algorithm in terms of both the distortion rate and the simplification rate. As shown in Fig 4, it can be seen that the distortion rate increases for both methods as the simplification rate increases. While, when the simplification rate is the same, the distortion rate of the method is significantly lower than that of the DP algorithm. The difference in distortion rate between the two methods also increases with increased trajectory simplification intensity. It showed that when the simplification requirement is lower than 87%, both methods can achieve the purpose of trajectory simplification and maintain the trajectory characteristics of the ship's original trajectory. However, our method obtained a better performance when the simplification rate is larger than 87%.

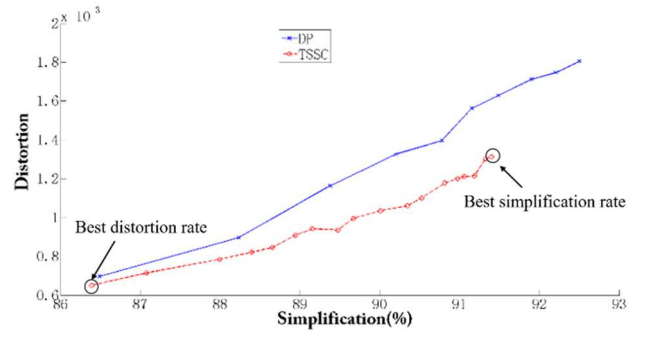
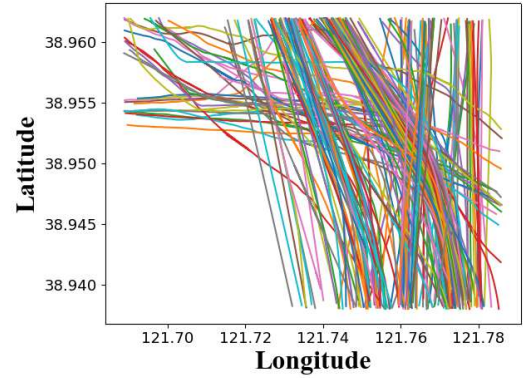


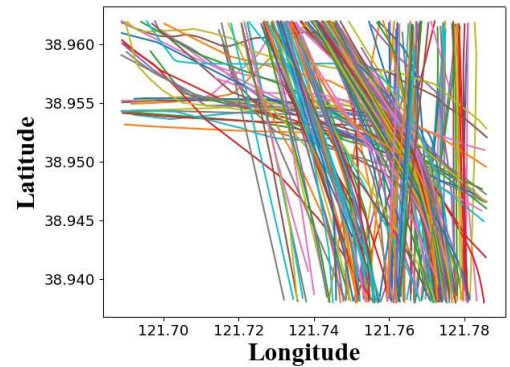
Fig. 4. Individual ship track

IV. REAL CASE OF TRAJECTORY SIMPLIFICATION

In order to validate the compression effect of our method, we implemented the proposed method on the AIS trajectory data of 15 vessels in Dalian port. The original trajectory was shown in Fig 5(a), and the compressed trajectory was shown in Fig 5(b). It can be seen from Fig 5 that there is no obvious trajectory information loss before and after trajectory data compression. The simplified compressed trajectories by our method can well retain the characteristics of the ship's motion and the original trajectories.



(a) Original Trajectory



(b) Simplified Trajectory

Fig. 5. Individual ship track

V. CONCLUSIONS

In this paper, a trajectory simplification method was proposed based on the trajectory segmentation and K-medoids clustering algorithm. The proposed method mainly consisted of two modules, namely trajectory segmentation and segment feature extraction. Firstly, the course and speed deviation threshold were given. The maneuver points were obtained

from these course and speed deviation threshold. Then, the trajectory was divided into segments in accordance with the maneuver points. Secondly, the K-medoids clustering algorithm was applied the extract feature of trajectory segments.

The experiment results showed a great performance on data of 15 ships and 232 trajectories in Dalian port. The compression and simplification capability were compared with the traditional DP algorithm. Compared with the DP algorithm, the distortion rate of the compressed trajectory is significantly reduced with the same compression intensity. Moreover, the algorithm maintains a good quality of simplification even under high intensity simplification requirements. The proposed method was sensitive for both global and local track features and can adapted to different tracks.

ACKNOWLEDGMENTS

This work is supported in part by the National Natural Science Foundation of China (under Grant Nos. 51939001, 61976033, U1813203, 61803064, 61751202); the Science and Technology Innovation Funds of Dalian (under Grant No. 2018J11CY022); the LiaoNing Revitalization Talents Program (under Grant Nos. XLYC1908018, XLYC1807046); the Natural Foundation Guidance Plan Project of Liaoning (2019-ZD-0151, 2020-HYLH-26); the Fundamental Research Funds for the Central Universities (under Grant No. 3132019345).

REFERENCES

- [1] Mou J M, Tak C V D, and Ligteringen H, "Study on collision avoidance in busy waterways by using AIS data," *Ocean Engineering*, 2010, 37(5-6):483-490.
- [2] Giuliana P , Michele V , and Karna B, "Vessel Pattern Knowledge Discovery from AIS Data: A Framework for Anomaly Detection and Route Prediction," *Entropy*, 2013, 15(6):2218-2245.
- [3] Altan Y C and Otay E N, "Spatial Mapping of Encounter Probability in Congested Waterways using AIS0 *Ocean Engineering*, 2018, 164(SEP.15):263-271.
- [4] International Maritime Organization(IMO), 1974, *International Convention for the Safety of Life at Sea (SOLAS)*.
- [5] Patroumpas K , Alevizos E ,and Artikis A , "Online Event Recognition from Moving Vessel Trajectories," 2016.
- [6] Qi, Le and Zhengyi Zheng, " Vessel trajectory data compression based on course alteration recognition," *Int. J. Simulat. Syst. Sci. Technol*, 17 (1) 5.1-5.8.2016.
- [7] Douglas, D.H and Peucker T.K, "Algorithms for the reduction of the number of points required to represent a digitized line or its caricature," *Cartographica: The International Journal for Geographic Information and Geovisualization* 10 (2) , 1973.
- [8] Muckell J, Hwang J.H, Lawson, C.T, and Ravi, S.S, "Algorithms for compressing GPS trajectory data: an empirical evaluation," In: *Sigspatial International Conference on Advances in Geographic Information Systems*, 402-405, 2010.
- [9] Liangbin Zhao and Guoyou Shi, "A method for simplifying ship trajectory based on improved Douglas-Peucker algorithm," *Ocean Engineering* 166 37-46, 2018.
- [10] M. Wang and Y. Dai, "Local Robust Feature Based on FAST Corner Detection," *Transactions of Beijing Institute of Technology*, vol. 33,no. 10, pp. 1045-1050, 2013.
- [11] Xie Y H , Han X W , and He Y G, "Information Technology in Multi-Object Tracking Based on Bilateral Structure Tensor Corner Detection for Mutual Occlusion," *Advanced Materials Research*, 2014, 977:502-506.
- [12] "Tracking based on Bilateral Structure Tensor Corner Detection for Mutual Occlusion". *Advanced Materials Research*, vol. 977, pp. 502-506, 2014.
- [13] Aggarwal C C, *Outlier Analysis*. Springer Publishing Company, Incorporated, 2015.